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Inventor(s)	Arntsen; John Frederick et al.

Optical switch for single and multiple projectors

Abstract

A projection system and method therefor comprises a first light source configured to emit a first-eye light, wherein the first-eye light includes a first set of wavelengths; a second light source configured to emit a second-eye light, wherein the second-eye light includes a second set of wavelengths; a first projector including first projection optics configured to receive a first input light; and an optical switch configured to be switched between an a first mode and a second mode, wherein the optical switch is configured to, in the first mode, combine the first-eye light and the second-eye light into a combined light and direct the combined light to the first projection optics as the first input light.

Inventors: Arntsen; John Frederick (Manchester-by-the-Sea, MA), Lippey; Barret (Foster City, CA)

Applicant: Dolby Laboratories Licensing Corporation (San Francisco, CA)

Family ID: 1000008818301

Assignee: Dolby Laboratories Licensing Corporation (San Francisco, CA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/608,563, filed on Nov. 3, 2021, which is the U.S. national stage entry of International Patent Application No. PCT/US2020/032248, filed on May 8, 2020, which claims priority to U.S. Provisional Patent Application No. 62/847,172, filed May 13, 2019, and U.S. Provisional Patent Application No. 62/845,263, filed May 8, 2019, all of which are incorporated herein by reference in their entirety.

BACKGROUND

1. Field of the Disclosure

(1) This application relates generally to image projection systems and methods.

2. Description of Related Art

(2) Digital projection systems typically utilize a light source and an optical system to project an image onto a surface or screen. The optical system may include components such as mirrors, lenses, waveguides, optical fibers, beam splitters, beam combiners, diffusers, spatial light modulators (SLMs), and the like.

(3) Some projection systems are capable of three-dimensional (3D) projection; that is, projecting an image onto a screen that can be perceived by a viewer in three dimensions. 3D systems may be designed for use with dual laser projectors using techniques including spectral separation. In a spectral separation system, the projector emits light to provide six primary colors (“primaries”) and the left-eye and right-eye images are separated through spectral filtering. This may be accomplished by providing the viewer with 3D glasses which include a triple band filter for each eye, such that each eye sees a different RGB spectrum. 3D systems may also be capable of two-dimensional (2D) projection in some modes; that is, projecting an image onto a screen that is perceived by a viewer in two dimensions.

(4) When using a 3D projection system to project 2D images, comparative examples utilize the dual projectors of the 3D projection system in either a stacked or side-by-side arrangement with some minimum distance between the projectors, and their projected images are overlaid at the screen. However, because there exists a minimum distance between the dual projectors of the comparative example systems and there are manufacturing variations between the optical components of the two projectors, such systems may suffer from image differences which prevent perfectly overlaying the two images.

BRIEF SUMMARY OF THE DISCLOSURE

(5) Various aspects of the present disclosure relate to projection systems and methods that may be optically switched between 3D projection and 2D projection modes and, when in the 2D projection modes, do not exhibit keystone artifacts.

(6) In one exemplary aspect of the present disclosure, there is provided a projection system comprising: a first light source configured to emit a first-eye light, wherein the first-eye light includes a first set of wavelengths; a second light source configured to emit a second-eye light, wherein the second-eye light includes a second set of wavelengths; a first projector including first projection optics configured to receive a first input light; and an optical switch configured to be switched between an a first mode and a second mode, wherein the optical switch is configured to, in the first mode, combine the first-eye light and the second-eye light into a combined light and direct the combined light to the first projection optics as the first input light.

(7) In another exemplary aspect of the present disclosure, there is provided a method of image projection comprising: emitting a first-eye light by a first light source, wherein the first-eye light includes a first set of wavelengths; emitting a second-eye light by a second light source, wherein the second-eye light includes a second set of wavelengths; receiving a first input light by a first projector including first projection optics; and switching an optical switch between an a first mode and a second mode, wherein the optical switch is configured to, in the first mode, combine the first-eye light and the second-eye light into a combined light and direct the combined light to the first

projection optics as the first input light.

(8) In this manner, various aspects of the present disclosure provide for the projection display of light using a system that may be switched between 2D operation and 3D operation, and that does not exhibit keystone artifacts in 2D operation. Thus, various aspects of the present disclosure effect improvements in at least the technical fields of image projection, cinematography, signal processing, and the like.

Description

DESCRIPTION OF THE DRAWINGS

(1) These and other more detailed and specific features of various embodiments are more fully disclosed in the following description, reference being had to the accompanying drawings, in which:

(2) FIG. 1 illustrates a block diagram of an exemplary projection system according to various aspects of the present disclosure;

(3) FIGS. 2A-2B illustrate exemplary projector details for the projection system according to FIG. 1;

(4) FIG. 3 illustrates a block diagram of another exemplary projection system according to various aspects of the present disclosure;

(5) FIGS. 4A-4B illustrate exemplary projector details for the projection system according to FIG. 3;

(6) FIG. 5 illustrates a block diagram of another exemplary projection system according to various aspects of the present disclosure;

(7) FIGS. 6A-6B illustrate exemplary projector details for the projection system according to FIG. 5; and

(8) FIG. 7 illustrates an exemplary device implementing various aspects of the present disclosure.

DETAILED DESCRIPTION

(9) This disclosure and aspects thereof can be embodied in various forms, including hardware or circuits controlled by computer-implemented methods, computer program products, computer systems and networks, user interfaces, and application programming interfaces; as well as hardware-implemented methods, signal processing circuits, memory arrays, application specific integrated circuits, field programmable gate arrays, and the like. The foregoing summary is intended solely to give a general idea of various aspects of the present disclosure, and does not limit the scope of the disclosure in any way.

(10) In the following description, numerous details are set forth, such as spectra, timings, operations, and the like, in order to provide an understanding of one or more aspects of the present disclosure. It will be readily apparent to one skilled in the art that these specific details are merely exemplary and not intended to limit the scope of this application.

(11) Moreover, while the present disclosure focuses mainly on examples in which the various elements are used in digital projection systems, it will be understood that this is merely one example of an implementation. It will further be understood that the disclosed systems and methods can be used in any device in which there is a need to project light; for example, cinema, consumer and other commercial projection systems, optical communications, heads-up displays, virtual reality displays, and the like.

(12) Projection Systems

(13) In some implementations, projection systems (e.g., certain digital cinema laser projection systems) use two projectors to selectively implement both 2D and stereoscopic 3D modes of operation. For example, such a 3D projection system includes two projectors that, when operating in a 3D mode, continuously project a left-eye image and a right-eye image each including three

primaries, respectively. In a 2D mode, the two projectors playback the same content and the images are overlaid at the screen, such that the resultant image includes six primaries.

(14) As noted above, comparative examples of 3D projection systems which include a 2D mode may utilize both projectors and overlay the respective images on a screen, thereby to enhance brightness and contrast. Because there is some minimum distance between the projection lenses of the two projectors in comparative example systems (vertically where the two projectors are stacked, or horizontally where the two projectors are side-by-side), such comparative example systems exhibit mirrored keystone artifacts (i.e., portions of the projected image that do not overlap, and thus are perceived by the viewer as a “double image”) in the 2D mode as a result of the distance between the two projectors. Also, even if the two projectors are matched as closely as possible, manufacturing variations in the projection lens and other optical parts may make differences between the two images that prevent an acceptable overlay.

(15) Comparative methods to remedy these defects may be complicated and/or costly. For example, the use of pixel warping to remedy the defects requires sophisticated optical hardware and custom software solutions. Although such systems may work under particular conditions, any projector movement will result in overlay defects reappearing. Even if the warping routine is performed on a regular basis in view of potential movement, the routine may take up to 30 minutes and requires projection systems to be warmed to operating temperature to ensure no subsequent thermal drift. In any event, pixel warping is strongly objected to by many movie studios and other content creators (especially during the movie mastering and grading process) because it alters the content.

(16) Projection systems according to various aspects of the present disclosure are described with respect to dual laser projectors implementing a spectral separation technique; however, the present disclosure may also be implemented using other light sources such as lamps. Because laser light sources are low-etendue sources, the light beams emitted therefrom can be efficiently formed into collimated beams and guided through various optical components of the projection system, such as the optical switch described in more detail below. Each laser projector may provide three of the six primaries. When operating in the 2D mode, all six primaries are projected onto the screen to create a single image that may be viewed by the viewer (e.g., an audience member in a theater) as a 2D image. When operating in the 3D mode, the sets of three primaries from each laser projector are projected onto the screen to create a left-eye image and a right-eye image, respectively, and the left-eye and right-eye images are separated through spectral filtering and separately provided to the left and right eyes of the viewer. In this manner, the viewer perceives the projected image as a 3D image.

(17) In order to view the projected image in three dimensions, the viewer may be provided with glasses having optical filters therein. For example, the glasses may have a left optical filter disposed over the viewer's left eye and configured to pass light intended for the left eye thereto while blocking light intended for the right eye, and may have a right optical filter disposed over the viewer's right eye and configured to pass light intended for the right eye thereto while blocking light intended for the left eye. To operate thusly, the left optical filter and the right optical filter have different acceptance bands. Differences in the image presented to the viewers left and right eyes cause the projected images to appear 3D.

(18) To eliminate the keystone artifacts or other deleterious effects that exist in comparative examples of 3D-capable projection systems in 2D operation, various aspects of the present disclosure include an optical switch that combines light from both emission sources into one projector. For 3D operation, the optical switch is configured such that the light from each emission source is respectively fed into its corresponding projector. The optical switch can either be a combiner or a splitter. Theaters that predominantly exhibit 2D content will benefit from reduced hours accumulated on the secondary projector because it will only be used for 3D operation.

(19) Optical Switch—Combiner

(20) For projection systems in which each set of lasers is delivered over independent fiber optics,

the optical switch is a combiner. This configuration allows all light from both left- and right-eye sets of lasers to be switched into one projector in a first mode (e.g., for 2D operation), or each set of laser outputs to be switched to each projector respectively in a second mode (e.g., for 3D operation). In an exemplary digital cinema laser projection system, the two sets of lasers have different nominal primary wavelengths (e.g., R1, G1, and B1 for one laser and R2, G2, and B2 for the other laser) with a wavelength separation of about 10-30 nm between corresponding primaries. A coating on a flat optic may be designed to reflect one set of RGB wavelengths and pass the other set of RGB wavelengths. With this combiner optic positioned in the beam path, the beams can be overlapped and combined before launching into the projection system for single-projector 2D operation. With the combiner optic moved out of the beam path, each RGB set is delivered appropriately to each projector for 3D operation. The movement of the combiner optic may be manual or automated on a rotating or translating mechanism. FIG. 1 illustrates an exemplary implementation of such a system.

(21) In particular, FIG. 1 illustrates a projection system **10** which includes a first laser light source **100L** which outputs a first set of wavelengths (e.g., three primaries) corresponding to left-eye light and a second laser light source **100R** which outputs a second set of wavelengths (e.g., three primaries) corresponding to right-eye light. The first laser light source **100L** is, for example, a first plurality of laser emitters (e.g., fiber lasers, laser diodes, or combinations thereof) which output light via an optical fiber which terminates in a first optical coupler **111**. Alternatively, each (or subsets) of the primaries corresponding to the left-eye light may be output via a separate optical fiber which terminates in the first optical coupler **111**, or in a plurality of first optical couplers **111**. The second laser light source **100R** is, for example, a second plurality of laser emitters (e.g., fiber lasers, laser diodes, or combinations thereof) which output light via a separate optical fiber which terminates in a second optical coupler **112**. Alternatively, each (or subsets) of the primaries corresponding to the right-eye light may be output via a separate optical fiber which terminates in the second optical coupler **112**, or in a plurality of second optical couplers **112**. The first laser light source **100L** and the second laser light source **100R** may be disposed in independent cabinets or may be consolidated into one cabinet. The projection system **10** includes a first projector **120L** which, in a 3D mode of the projection system, projects a left-eye image. The projection system **10** further includes a second projector **120R**. In the 3D mode, the second projector **120R** projects a right-eye image. In a 2D mode, the second projector **120R** projects all six primaries as a single image.

(22) Both the first optical coupler **111** and the second optical coupler **112** are optically connected to the second projector **120R**. Light from the first optical coupler **111** and the second optical coupler **112** respectively follow a first optical path **131** and a second optical path **132** to a first optical system **121**, which selectively operates as an optical switch and right-eye filter. Depending on the state of the optical switch, the first optical system **121** may direct a portion of the received light to a third optical coupler **122** for further optical processing as will be described in more detail below. Moreover, depending on the state of the optical switch, the first optical system **121** may direct all or a portion of the received light via a third optical path **141** to first projection optics **123**. The first projection optics **123** project an image via a first output optical path **151**.

(23) When the state of the optical switch is such that a portion of the received light is directed to the third optical coupler **122**, the light is transported via an optical fiber to a fourth optical coupler **125** associated with the second projector **120L**. The optical fiber may be a single fiber or a multi-fiber bundle. The second projector **120L** includes a second optical system **124**, which operates as a left-eye filter. Light from the second optical system **124** is directed via a fourth optical path **142** to second projection optics **126**. The second projection optics **126** project an image via a second output optical path **152**. The first projection optics **123** and the second projection optics **126** include optical components such as light homogenizers (e.g., integrating rods, fly's-eye optical components, and combinations thereof), lenses (e.g., Fourier transform lenses, zoom lens arrays,

projection lenses, and combinations thereof), mirrors, SLMs (e.g., digital micromirror devices (DMDs)), and so on.

(24) When the projection system **10** is in a 2D mode, the optical switch of the first optical system **121** is configured to direct all six primaries to the first projection optics **123**. Thus, the image output via the first output optical path **151** corresponds to a six-primary 2D image. When the projection system **10** is in a 3D mode, the optical switch of the first optical system **121** is configured to direct the three primaries emitted by the first laser light source **100L** to the second projection optics **126** and to direct the three primaries emitted by the second laser light source **100R** to the first projection optics **123**. Thus, the image output via the second output optical path **152** corresponds to a three-primary left-eye image and the image output via the first output optical path **151** corresponds to a three-primary right-eye image.

(25) FIGS. 2A-2B illustrate the modes of the projection system **10** in greater detail. In particular, FIG. 2A illustrates a part of the projection system **10** in the 2D mode, and FIG. 2B illustrates the part of the projection system **10** in the 3D mode. As illustrated in FIGS. 2A-2B, the first optical system **121** includes a first lens **201**, a second lens **202**, a third lens **203**, a fourth lens **204**, a movable mirror **211** (or other reflective element), and a first filter **212**. The second optical system **124** includes a fifth lens **205**, a second filter **213**, and a sixth lens **206**. In some examples, the first filter **212** and the second filter **213** are trim filters that reflect unwanted tails of the emission peaks. The movable mirror **211** is movable between a position in the beam path and a position out of the beam path. The movement of the movable mirror **211** may be manual or automated on a rotating or translating mechanism.

(26) When the projection system **10** is in the 2D mode illustrated in FIG. 2A, the movable mirror **211** is located in an optical path of light received via the first optical coupler **111** through the first lens **201**. In this mode, the movable mirror **211** reflects the light from the first optical coupler **111** toward the first filter **212**. The first filter **212** acts both as a combiner and as a trim filter. The first filter **212** is located in an optical path of light received via the second optical coupler **112** through the second lens **202** and reflects a portion of the light corresponding to unwanted light (e.g., unwanted tails of the emission bands) from the second laser light source **100R** while transmitting a portion of the light corresponding to the desired light from the second laser light source **100R**. While the first filter **212** is illustrated as a relatively-thick element, in practice the first filter **212** (and/or other filters illustrated and described herein) may be sufficiently thin that the transmitted and reflected beams are substantially overlapping, thereby to improve uniformity. The first filter **212** is also located in an optical path of light reflected by the movable mirror **211** in the 2D mode, such that the transmitted right-eye light and the doubly-reflected left-eye light travel along the same (or substantially the same) optical path through the fourth lens **204** and to the first projection optics **123**. As illustrated, this optical path corresponds to the third optical path **141** of FIG. 1. Thus, in the 2D mode, the first projection optics **123** output an image along the first output optical path **151** which corresponds to a six-primary 2D image exhibiting reduced or no overlay artifacts.

(27) When the projection system **10** is in the 3D mode illustrated in FIG. 2B, the movable mirror **211** is located outside of the optical path of light received via the first optical coupler **111** through the first lens **201**. Thus, the light from the first optical coupler **111** proceeds through the third lens **203** and into the third optical coupler **122**, where it is transported via an optical fiber to the fourth optical coupler **125** associated with the second projector **120L**. The light from the second optical coupler **112** passes through the second lens **202** to the first filter **212**, where the portion of the light corresponding to unwanted light (e.g., unwanted tails of the emission bands) from the second laser light source **100R** is reflected and the portion of the light corresponding to the desired light from the second laser light source **100R** is transmitted through the fourth lens **204** and to the first projection optics **123**.

(28) In the second projector **120L**, the light input by the fourth optical coupler **125** passes through the fifth lens **205** to the second filter **213**. The second filter **213** is a trim filter that reflects a portion

of the light corresponding to unwanted tails of the emission bands from the first laser light source **100L** while transmitting a portion of the light corresponding to the desired peaks from the first laser light source **100L**. The transmitted portion passes through the sixth lens **206** and travels along an optical path (as illustrated, corresponding to the fourth optical path **142** of FIG. **1**) to the second projection optics **126**. Thus, in the 3D mode the first projection optics **123** output an image along the first output optical path **151** which corresponds to a three-primary right-eye image and the second projection optics **126** output an image along the second output optical path **152** which corresponds to a three-primary left-eye image.

(29) The combiner configuration of the optical switch allows left and right trim filters (i.e., the first filter **212** and the second filter **213**) to be present in both 2D and 3D operation. In 2D operation, this ensures defined spectra to meet color gamut specifications, even with lasers that may contain emission that is out of band. In 3D operation, this eliminates unwanted spectral bands from each projector respectively, and thereby reduces crosstalk between the eyes.

(30) While FIGS. **2A-2B** illustrate the optical switch as combining the left-eye and right-eye light by reflecting the left-eye light and combining it with the right-eye light in the 2D mode, the present disclosure is not so limited. In some implementations, the movable mirror **211** and the first filter **212** may be reversed, and appropriate adjustments may be made to the angle and axial position of the remaining optical components, so that the optical switch reflects the right-eye light and combines it with the left-eye light in the 2D mode. By locating the combiner inside the projector **120R**, there is no need to relaunch light from the combiner into a fiber patch cable. However, while FIGS. **2A-2B** illustrate the combiner as being internal to the second projector **120R**, the present disclosure is not so limited. In some implementations, the combiner may be externally located with fiber patch cables to each projector. The fiber patch cables between an external switch and the projectors may be selected to accommodate different distances between the switch and projectors, thereby allowing the distance from the lasers to the external switch to be made a standard length.

(31) Moreover, while FIGS. **1-2B** illustrate the first projector **120L** and the second projector **120R** using separate blocks, the present disclosure is not limited to implementations where the first projector **120L** and the second projector **120R** are confined to separate cabinets or casings. In some implementations, the first projector **120L** and the second projector **120R** may be included in a single projector cabinet from which the first projection optics **123** and (in the 3D mode) the second projection optics **126** output their respective images, the fiber patch cable between the third optical coupler **122** and the fourth optical coupler **125** (as well as the couplers themselves) may be eliminated.

(32) Optical Switch—Splitter

(33) For projection systems in which both sets of lasers are delivered over a single fiber optic, the optical switch may be a splitter. This configuration allows all light from both left- and right-eye sets of lasers to be switched into one projector in a first mode (e.g., for 2D operation), or each set of laser outputs to be switched to each projector respectively in a second mode (e.g., for 3D operation). As noted above, in an exemplary digital cinema laser projection system, the two sets of lasers have different nominal primary wavelengths (e.g., R1, G1, and B1 for one laser and R2, G2, and B2 for the other laser) with a wavelength separation of about 10-30 nm between corresponding primaries. A coating on a flat optic may be designed to reflect one set of RGB wavelengths and pass the other set of RGB wavelengths. With this splitter optic positioned in the beam path, one set of wavelengths is passed to one projector and the other set of wavelengths is reflected to the other projector for two-projector 3D operation. With the splitter optic moved out of the beam path, both sets of RGB wavelengths are delivered to one projector for single-projector 2D operation. The movement of the splitter optic may be manual or automated on a rotating or translating mechanism. FIG. **3** illustrates an exemplary implementation of such a system.

(34) In particular, FIG. **3** illustrates a projection system **30** which includes a laser light source **300** which outputs a combined set of wavelengths (e.g., six primaries) corresponding to both a first

subset of wavelengths for left-eye light and a second subset of wavelengths for right-eye light. The laser light source **300** is, for example, a plurality of laser emitters (e.g., fiber lasers, laser diodes, or combinations thereof) which output light via an optical fiber which terminates in a first optical coupler **310**. Alternatively, each (or subsets) of the six primaries may be output via a separate optical fiber which terminates in the first optical coupler **310**, or in a plurality of first optical couplers **310**. The projection system **50** includes a first projector **320L** which, in a 3D mode of the projection system, projects a left-eye image. The projection system **30** further includes a second projector **320R**. In the 3D mode, the second projector **320R** projects a right-eye image. In a 2D mode, the second projector **320R** projects all six primaries a single image.

(35) The optical coupler **310** is optically connected to the second projector **320R**. Light from the optical coupler **310** follows a first optical path **330** to a first optical system **321**, which selectively operates both as an optical switch and as a right-eye filter. Depending on the state of the optical switch, the first optical system **321** may direct a portion of the received light to a second optical coupler **322** for further optical processing as will be described in more detail below. Moreover, depending on the state of the optical switch, the first optical system **321** may direct all or a portion of the received light via a second optical path **341** to first projection optics **323**. The first projection optics **323** project an image via a first output optical path **351**.

(36) When the state of the optical switch is such that a portion of the received light is directed to the second optical coupler **322**, the light is transported via an optical fiber to a third optical coupler **325** associated with the second projector **320L**. The optical fiber may be a single fiber or a multi-fiber bundle. The second projector **320L** includes a second optical system **324**, which operates as a left-eye filter. Light from the second optical system **324** is directed via a third optical path **342** to second projection optics **326**. The second projection optics **326** project an image via a second output optical path **352**. The first projection optics **323** and the second projection optics **326** include optical components such as light homogenizers (e.g., integrating rods, fly's-eye optical components, and combinations thereof), lenses (e.g., zoom lens arrays, projection lenses, and combinations thereof), mirrors, SLMs (e.g., DMDs), and so on.

(37) When the projection system **30** is in a 2D mode, the optical switch of the first optical system **321** is configured to direct all six primaries to the first projection optics **323**. Thus, the image output via the first output optical path **351** corresponds to a six-primary 2D image. When the projection system **30** is in a 3D mode, the optical switch of the first optical system **321** is configured to direct the three primaries corresponding to the left-eye image emitted by the laser light source **300** to the second projection optics **326** and to direct the three primaries corresponding to the right-eye image emitted by the laser light source **300** to the first projection optics **323**. Thus, the image output via the second output optical path **352** corresponds to a three-primary left-eye image and the image output via the first output optical path **351** corresponds to a three-primary right-eye image.

(38) FIGS. **4A-4B** illustrate the modes of the projection system **30** in greater detail. In particular, FIG. **4A** illustrates a part of the projection system **30** in the 2D mode, and FIG. **4B** illustrates the part of the projection system **30** in the 3D mode. As illustrated in FIGS. **4A-4B**, the first optical system **321** includes a first lens **401**, a second lens **402**, a movable first filter **411**, and a mirror **412** (or other reflective element). The second optical system **324** includes a third lens **403**, a second filter **413**, and a fourth lens **404**. In some examples, the first filter **411** and the second filter **413** are trim filters that reflect unwanted tails of the emission peaks. The first filter **411** is movable between a position in the beam path and a position out of the beam path. The movement of the first filter **411** may be manual or automated on a rotating or translating mechanism.

(39) When the projection system **30** is in the 2D mode illustrated in FIG. **4A**, the first filter **411** is located outside of the optical path of light received via the first optical coupler **310** through the first lens **401**. Thus, all (or substantially all) of the light from the first optical coupler **310** proceeds along the second optical path **341** and to the first projection optics **423**. Thus, in the 2D mode, the

first projection optics **423** output an image along the first output optical path **351** which corresponds to a six-primary 2D image exhibiting reduced or no overlay artifacts.

(40) When the projection system **30** is in the 3D mode illustrated in FIG. **4B**, the first filter **411** is located in the optical path of light received via the optical coupler **310** through the first lens **401**. The first filter **411** acts as a splitter and as a trim filter. The first filter reflects a portion of the light corresponding to the left-eye image emitted by the laser light source **300** and a portion of the light corresponding to unwanted tails of the emission bands for the right-eye image emitted by the laser light source **300**, while transmitting a portion of the light corresponding to the desired peaks of the right-eye image from the laser light source **300**. In the particular example illustrated in FIG. **4B**, the mirror **412** is located in an optical path of the reflected portion of light, and reflects the incident light through the second lens **402** to the second optical coupler **322**, where it is transported via an optical fiber to the third optical coupler **325** associated with the second projector **320L**. In other examples, the mirror **412** may be omitted, in which case the second optical coupler **322** may be repositioned such that it lies in the optical path of the reflected portion of light.

(41) In the second projector **320L**, the light input by the third optical coupler **325** passes through the third lens **403** to the second filter **413**. The second filter **413** is a trim filter that reflects a portion of the light corresponding to unwanted light of both the left-eye and right-eye emission bands from the laser light source **300** while transmitting a portion of the light corresponding to the desired peaks of the left-eye image from the laser light source **300**. The transmitted portion passes through the fourth lens **404** and travels along an optical path (as illustrated, corresponding to the fourth optical path **342** of FIG. **3**) to the second projection optics **326**. Thus, in the 3D mode the first projection optics **323** output an image along the first output optical path **351** which corresponds to a three-primary right-eye image and the second projection optics **326** output an image along the second output optical path **352** which corresponds to a three-primary left-eye image.

(42) In the particular illustration of FIGS. **4A-4B**, refraction of light transmitted through the first filter **411** may cause the optical axis of the second optical path **341** to shift laterally. In the event that such a shift negatively affects the performance of the first projection optics **323**, one or more optical elements such as an electronic crystal may be disposed between the first filter **411** and the first projection optics **323** to provide an equal and opposite lateral shift to the optical axis of the second optical path **341**. Where such optical elements are used, they may be configured to move along with the first filter **411** so that they do not affect light in the 2D mode.

(43) While FIGS. **4A-4B** illustrate the optical switch as splitting the left-eye and right-eye light such that the left-eye light is sent to a separate projector in the 3D mode, the present disclosure is not so limited. In some implementations, the filtering and reflection wavelengths of the first filter **411** may be appropriately chosen such that the right-eye light is sent to a separate projector in the 3D mode. By locating the splitter inside the projector **120R**, there is no need to relaunch light from the splitter into a fiber patch cable. However, while FIGS. **4A-4B** illustrate the splitter as being internal to the second projector **420R**, the present disclosure is not so limited. In some implementations, the splitter may be externally located with fiber patch cables to each projector. The fiber patch cables between an external switch and the projectors may be selected to accommodate different distances between the switch and projectors, thereby allowing the distance from the lasers to the external shift to be made a standard length.

(44) Moreover, while FIGS. **3-4B** illustrate the first projector **320L** and the second projector **320R** using separate blocks, the present disclosure is not limited to implementations where the first projector **320L** and the second projector **320R** are confined to separate cabinets or casings. In some implementations, the first projector **320L** and the second projector **320R** may be included in a single projector cabinet from which the first projection optics **323** and (in the 3D mode) the second projection optics **326** output their respective images, the fiber patch cable between the third optical coupler **322** and the fourth optical coupler **325** (as well as the couplers themselves) may be

eliminated.

(45) Optical Switch—Splitter/Recombiner

(46) For projection systems in which both sets of lasers are delivered over a single fiber optic, the optical switch may alternatively be a splitter/recombiner. This configuration similarly allows all light from both left- and right-eye sets of lasers to be switched into one projector in a first mode (e.g., for 2D operation), or each set of laser outputs to be switched to each projector respectively in a second mode (e.g., for 3D operation); however, this configuration may result in a reduced or eliminated launch offset (lateral shift) as compared to the splitter configuration discussed above. Moreover, this configuration may result in eased manufacturing tolerances and/or reduced precision requirements for the positioning of various components.

(47) As noted above, in an exemplary digital cinema laser projection system, the two sets of lasers have different nominal primary wavelengths (e.g., R1, G1, and B1 for one laser and R2, G2, and B2 for the other laser) with a wavelength separation of about 10-30 nm between corresponding primaries. A coating on a flat optic may be designed to reflect one set of RGB wavelengths and pass the other set of RGB wavelengths. With a recombiner portion of the splitter/recombiner optic positioned in the beam path, one set of wavelengths is passed to one projector and the other set of wavelengths is reflected to the other projector for two-projector 3D operation. With the recombiner portion of the splitter/recombiner optic moved out of the beam path, both sets of RGB wavelengths are delivered to one projector for single-projector 2D operation. The movement of the recombiner portion may be manual or automated on a rotating or translating mechanism. FIG. 5 illustrates an exemplary implementation of such a system.

(48) In particular, FIG. 5 illustrates a projection system **50** which includes a laser light source **500** which outputs a combined set of wavelengths (e.g., six primaries) corresponding to both a first subset of wavelengths for left-eye light and a second subset of wavelengths for right-eye light. The laser light source **500** is, for example, a plurality of laser emitters (e.g., fiber lasers, laser diodes, or combinations thereof) which output light via an optical fiber which terminates in a first optical coupler **510**. Alternatively, each (or subsets) of the six primaries may be output via a separate optical fiber which terminates in the first optical coupler **510**, or in a plurality of first optical couplers **510**. The projection system **50** includes a first projector **520L** which, in a 3D mode of the projection system, projects a left-eye image. The projection system **50** further includes a second projector **520R**. In the 3D mode, the second projector **520R** projects a right-eye image. In a 2D mode, the second projector **520R** projects all six primaries a single image.

(49) The optical coupler **510** is optically connected to the second projector **520R**. Light from the optical coupler **510** follows a first optical path **530** to a first optical system **521**, which selectively operates both as an optical switch and as a right-eye filter. Depending on the state of the optical switch, the first optical system **521** may direct a portion of the received light to a second optical coupler **522** for further optical processing as will be described in more detail below. Moreover, depending on the state of the optical switch, the first optical system **521** may direct all or a portion of the received light via a second optical path **541** to first projection optics **523**. The first projection optics **523** project an image via a first output optical path **551**.

(50) When the state of the optical switch is such that a portion of the received light is directed to the second optical coupler **522**, the light is transported via an optical fiber to a third optical coupler **525** associated with the second projector **520L**. The optical fiber may be a single fiber or a multi-fiber bundle. The second projector **520L** includes a second optical system **524**, which operates as a left-eye filter. Light from the second optical system **524** is directed via a third optical path **542** to second projection optics **526**. The second projection optics **526** project an image via a second output optical path **552**. The first projection optics **523** and the second projection optics **526** include optical components such as light homogenizers (e.g., integrating rods, fly's-eye optical components, and combinations thereof), lenses (e.g., zoom lens arrays, projection lenses, and combinations thereof), mirrors, SLMs (e.g., DMDs), and so on.

(51) When the projection system **50** is in a 2D mode, the optical switch of the first optical system **521** is configured to direct all six primaries to the first projection optics **523**. Thus, the image output via the first output optical path **551** corresponds to a six-primary 2D image. When the projection system **50** is in a 3D mode, the optical switch of the first optical system **521** is configured to direct the three primaries corresponding to the left-eye image emitted by the laser light source **500** to the second projection optics **526** and to direct the three primaries corresponding to the right-eye image emitted by the laser light source **500** to the first projection optics **523**. Thus, the image output via the second output optical path **552** corresponds to a three-primary left-eye image and the image output via the first output optical path **551** corresponds to a three-primary right-eye image.

(52) FIGS. **6A-6B** illustrate the modes of the projection system **50** in greater detail. In particular, FIG. **6A** illustrates a part of the projection system **50** in the 2D mode, and FIG. **6B** illustrates the part of the projection system **50** in the 3D mode. As illustrated in FIGS. **6A-6B**, the first optical system **521** includes a first lens **601**, a second lens **602**, a third lens **603**, a first filter **611**, a second filter **612**, a stationary mirror **613** (or other reflective element), and a movable mirror **614**. The second optical system **524** includes a fourth lens **604**, a fifth lens **605**, and a third filter **615**. In some examples, the first filter **611**, the second filter **612**, and the third filter **615** are trim filters that reflect unwanted tails of the emission peaks. The movable mirror **614** is movable between a position in a section of the beam path and a position out of the section of the beam path. The movement of the movable mirror **614** may be manual or automated on a rotating or translating mechanism.

(53) Light received from the first optical coupler **510** through the first lens **601** is incident on the first filter **611**. The first filter **611** reflects a portion of the light corresponding to the left-eye image emitted by the laser light source **500**, while transmitting a portion of the light corresponding to the desired peaks of the right-eye image from the laser light source **500** toward a first surface of the second filter **612**. The second filter **612** reflects a portion of the light that has passed through the first filter **611** corresponding to unwanted tails of the emission bands for the right-eye image emitted by the laser light source **500**, while transmitting a portion of the light corresponding to the desired peaks of the right-eye image from the laser light source **500**.

(54) Light reflected by the first filter **611** is incident on the stationary mirror **613**. When the projection system **50** is in the 2D mode illustrated in FIG. **6A**, the movable mirror **614** is located in the optical path of light reflected from the stationary mirror **613**. In this mode, the movable mirror **614** reflects the light reflected from the stationary mirror **613** toward a second surface of the second filter **612** opposite the first surface. The second filter **612** reflects the light reflected from the movable mirror **614** toward the second lens **602**, such that the transmitted right-eye light and the triply-reflected left-eye light travel along the same (or substantially the same) optical path through the second lens **602** and to the first projection optics **523**. Because these optical paths are the same (or substantially the same), the first optical system **521** corrects for lateral shift. As illustrated, this optical path corresponds to the third optical path **541** of FIG. **5**. Thus, in the 2D mode, the first projection optics **523** output an image along the first output optical path **551** which corresponds to a six-primary 2D image exhibiting reduced or no overlay artifacts or lateral shift.

(55) When the projection system **50** is in the 3D mode illustrated in FIG. **6B**, the movable mirror **614** is located outside of the optical path of the light reflected by the stationary mirror **613**. In this mode, the light reflected by the stationary mirror **613** proceeds through the third lens **603** and into the second optical coupler **522**, where it is transported via an optical fiber to the fourth optical coupler **525** associated with the second projector **520L**. In the second projector **520L**, the light input by the fourth optical coupler **525** passes through the fourth lens **604** to the third filter **615**. The third filter is a trim filter that reflects a portion of the light corresponding to unwanted tails of the emission bands of from the laser light source **500** while transmitting a portion of the light corresponding to the desired peaks from the laser light source **500**. The transmitted portion passes

through the fifth lens **605** and travels along an optical path (as illustrated, corresponding to the fourth optical path **542** of FIG. 5) to the second projection optics **526**.

(56) Meanwhile, the light transmitted through the first filter **611** (and not reflected toward the stationary mirror **613**) is incident on the first surface of the second filter **612**. As in the 2D mode, the second filter **612** reflects a portion of the light that has passed through the first filter **611** corresponding to unwanted tails of the emission bands for the right-eye image emitted by the laser light source **500**, while transmitting a portion of the light corresponding to the desired peaks of the right-eye image from the laser light source **500** through the second lens **602** and to the first projection optics **523**. Thus, in the 3D mode the first projection optics **523** outputs an image along the first output optical path **551** which corresponds to a three-primary right-eye image and the second projection optics output an image along the second output optical path **552** which corresponds to a three-primary left-eye image.

(57) While FIG. 6B illustrates the second filter **612** as reflecting the portion of the light that has passed through the first filter **611** corresponding to unwanted tails of the emission bands for the right-eye image emitted by the laser light source **500**, the present disclosure is not so limited. In other implementations, the first filter **611** reflects the unwanted tails of the emission bands for the right-eye image emitted by the laser light source **500** in addition to the portion of the light corresponding to the left-eye image emitted by the laser light source **500**, such that the third filter **615** reflects a portion of the light corresponding to unwanted light of both the left-eye and right-eye emission bands from the laser light source **500** while the projection system **50** is in the 3D mode.

(58) Furthermore, while FIGS. 6A-6B illustrate the optical switch as splitting the left-eye and right-eye light such that the left-eye light is sent to a separate projector in the 3D mode, the present disclosure is not so limited. In some implementations, the filtering and reflection wavelengths of the first filter **611** or the second filter **612** may be appropriately chosen such that the right-eye light is sent to a separate projector in the 3D mode. By locating the splitter/recombiner inside the projector **520R**, there is no need to relaunch light from the splitter/recombiner into a fiber patch cable. However, while FIGS. 6A-6B illustrate the splitter/recombiner as being internal to the second projector **520R**, the present disclosure is not so limited. In some implementations, the splitter/recombiner may be externally located with fiber patch cables to each projector. The fiber patch cables between an external switch and the projectors may be selected to accommodate different distances between the switch and projectors, thereby allowing the distance from the lasers to the external shift to be made a standard length.

(59) Moreover, while FIGS. 5-6B illustrate the first projector **520L** and the second projector **520R** using separate blocks, the present disclosure is not limited to implementations where the first projector **520L** and the second projector **520R** are confined to separate cabinets or casings. In some implementations, the first projector **520L** and the second projector **520R** may be included in a single projector cabinet from which the first projection optics **523** and (in the 3D mode) the second projection optics **526** output their respective images, the fiber patch cable between the third optical coupler **522** and the fourth optical coupler **525** (as well as the couplers themselves) may be eliminated.

(60) Device Implementation

(61) Projection systems including the above-described optical switch may be implemented by way of an electronic device including several optical and electronic components. FIG. 7 illustrates one such device **700**.

(62) As illustrated in FIG. 7, the device **700** includes a projection system **710**, a controller **720**, a memory **730**, and communications and I/O circuitry **740**. The projection system **710**, the controller **720**, the memory **730**, and the communications and I/O circuitry **740** communicate via a bus **750**. The projection system **710** may be, for example, the projection system **10** illustrated in FIG. 1, the projection system **30** illustrated in FIG. 3, or the projection system **50** illustrated in FIG. 5. The controller **720** may be, for example, one or more processors or other control circuitry, such as a

central processing unit (CPU), a graphics processing unit (GPU), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), and the like. The memory **730** may be, for example, a hard disk, a removable media drive, an optical or magnetic storage device, and combinations thereof. The memory **730** may include read-only memory (ROM) and/or random-access memory (RAM). The communications and I/O circuitry **740** may be, for example, a set of circuits, ports, connectors, etc., to allow the device **700** to communicate with a user or another device. The communications and I/O circuitry **740** may include wired communication interfaces and connect to user interface devices such as a mouse, a keyboard, a display, a touch screen, a wired network, and so on; additionally or alternatively, the communications and I/O circuitry **740** may include wireless communication interfaces such as Bluetooth, near-field communications (NFC), wireless local area networks (WLANs), and so on. The communications and I/O circuitry **740** may allow for the receipt of user instructions, streaming image data, device calibration, software or firmware updates, and so on.

(63) The memory **730** may store non-transitory computer-readable media storing instructions that, when executed by the controller **720**, cause the device to perform various operations. For example, the controller **720** may provide control signals to the projection system **710** (or to one or more actuators associated therewith) to cause the projection system to switch between a 2D mode (e.g., as illustrated in FIG. 2A, 4A, or 6A) and a 3D mode (e.g., as illustrated in FIG. 2B, 4B, or 6B). The controller **720** may provide control signals to the projection system **710** to control the light-emitting elements thereof to emit light in accordance with image data stored in the memory **730** and/or received via communications and I/O circuitry **740**. Additionally or alternatively, the controller **720** may provide control signals to one or more SLMs associated with the projection system **710** to modify emitted light in accordance with the image data, thereby to form a projected image on a screen **760**.

(64) Moreover, while FIG. 7 illustrates a single device **700** including all of the above-noted elements, in some implementations a portion or all of the projection system **710** may be separate from the device **700** (e.g., in a separate housing in communication with the device **700**). In one example, the portions of the projection system **510** associated with the emission of light (e.g., the first laser light source **100L** and second laser light source **100R** of FIG. 1, the laser light source **300** of FIG. 3, or the laser light source **500** of FIG. 5) are disposed in a common housing with device **700** whereas the optically-downstream components (e.g., the projectors **120L** and **120R** of FIG. 1, the projectors **320L** and **320R** of FIG. 3, or the projectors **520L** and **520R** of FIG. 5) are provided in a separate housing or housings in optical and/or electronic communication with the device **700**. In another example, the entirety of the projection system **710** is disposed in a separate housing in optical and/or electronic communication with the device **700**.

(65) Within the projection system **710**, various optical fibers may be used to direct light in one or more of the manners described above. In some examples, the dimensions of the individual optical fibers may be selected so as to reduce or eliminate path differences between various optical paths. For example, light may be transported within and/or among the projectors by various lengths of 50-800 μm fibers.

(66) Effects and Applications

(67) The above-described optical switch configurations, whether in a combiner implementation or a splitter implementation, results in the overlapping of the beams before launching into the projection optics. This ensures uniform illumination. The left-eye RGB wavelength set and the right-eye RGB wavelength set fiber outputs may be spatially combined, and there is no requirement that the outputs overlap before launching into the projector, although such a configuration may introduce some decrease in illumination uniformity. Moreover, the spatial overlapping of the beams allows for sets of lasers with identical spectra to be combined. Two or more sets of right or left-eye wavelength sets can be combined to create higher power configurations.

(68) The above description has been particularly described with respect to the case where the

projection system operates using a spectral separation technique. That is, the above projection systems utilize a left laser source which emits light with three peaks (one in the red wavelength range, one in the green wavelength range, and one in the blue wavelength range) and a right laser light source which also emits light with three peaks (one in the red wavelength range, one in the green wavelength range, and one in the blue wavelength range), where the peaks in light corresponding to the left eye are spectrally separated from the peaks in light corresponding to the right eye. In other words, the peaks within each wavelength range generated by the left laser light source may be shorter in wavelength than the peaks within each wavelength range generated by the right laser light source, or vice versa. However, the present disclosure is not so limited.

(69) In some examples, the projection systems described herein may operate using a technique other than spectral separation. In one example, the projection systems may operate instead using a polarization separation technique, in which left-eye light is polarized in a first direction and right-eye light is polarized in a second direction perpendicular to the first direction or in which left-eye light is circularly polarized in a clockwise direction and right-eye light is circularly polarized in a counterclockwise direction (or vice versa).

(70) The systems and methods described herein are comparatively simple and, by design, eliminate any double image artifact (i.e., keystone artifacts) resulting from two-projector 2D operation. Moreover, the present disclosure may reduce or eliminate the need for periodic alignment adjustments due to minor mechanical or thermal movements that would be present in comparative example systems.

(71) Systems and devices in accordance with the present disclosure may take any one or more of the following configurations.

(72) (1) A projection system, comprising: a first light source configured to emit a first-eye light, wherein the first-eye light includes a first set of wavelengths; a second light source configured to emit a second-eye light, wherein the second-eye light includes a second set of wavelengths; a first projector including first projection optics configured to receive a first input light; and an optical switch configured to be switched between an a first mode and a second mode, wherein the optical switch is configured to, in the first mode, combine the first-eye light and the second-eye light into a combined light and direct the combined light to the first projection optics as the first input light.

(73) (2) The projection system according to (1), further comprising: a second projector including second projection optics configured to receive a second input light, wherein the optical switch is configured to, in the second mode, direct the first-eye light to the second projection optics as the second input light.

(74) (3) The projection system according to (2), wherein the second projection optics includes a spatial light modulator configured to perform a spatial modulation on the second input light in response to an image data, thereby to generate a first-eye projection image when the optical switch is in the second mode.

(75) (4) The projection system according to any one of (1) to (3), wherein the optical switch is configured to, in the second mode, direct the second-eye light to the first projection optics as the first input light.

(76) (5) The projection system according to any one of (1) to (4), wherein the optical switch includes a reflective element and a first filter, and when the optical switch is in the first mode: the reflective element is configured to reflect the first-eye light to the first filter as first-eye reflected light, and the first filter is configured to reflect a first portion of the second-eye light, to transmit a second portion of the second-eye light as a first portion of the combined light, and to reflect the first-eye reflected light as a second portion of the combined light.

(77) (6) The projection system according to (5), further comprising: a second projector including second projection optics configured to receive a second input light, wherein, when the optical switch is in the second mode: the reflective element is configured to allow the first-eye light to pass to the second projector.

(78) (7) The projection system according to (6), wherein the second projector includes a second filter configured to, when the optical switch is in the second mode, reflect a first portion of the first-eye light and transmit a second portion of the first-eye light as the second input light.

(79) (8) The projection system according to any one of (1) to (7), wherein the first projection optics includes a spatial light modulator configured to perform a spatial modulation on the first input light in response to an image data, thereby to generate a second-eye projection image when the optical switch is in the second mode and to generate a combined projection image when the optical switch is in the first mode.

(80) (9) The projection system according to any one of (1) to (4), wherein the optical switch includes a first filter and a reflective element, and when the optical switch is in the first mode: the first filter is configured to allow the first-eye light and the second-eye light to pass to the first projection optics as the combined light.

(81) (10) The projection system according to (9), further comprising: a second projector including second projection optics configured to receive a second input light, wherein, when the optical switch is in the second mode: the first filter is configured to reflect the first-eye light and a first portion of the second-eye light to the reflective element as first-eye reflected light, and to transmit a second portion of the second-eye light as the first input light, and the reflective element is configured to direct the first-eye reflected light to the second projector.

(82) (11) The projection system according to (10), wherein the second projector includes a second filter configured to, when the optical switch is in the second mode, reflect a first portion of the first-eye reflected light and transmit a second portion of the first-eye reflected light as the second input light.

(83) (12) The projection system according to any one of (1) to (11), wherein the first light source and the second light source are laser light sources.

(84) (13) A method of image projection comprising: emitting a first-eye light by a first light source, wherein the first-eye light includes a first set of wavelengths; emitting a second-eye light by a second light source, wherein the second-eye light includes a second set of wavelengths; receiving a first input light by a first projector including first projection optics; and switching an optical switch between an a first mode and a second mode, wherein the optical switch is configured to, in the first mode, combine the first-eye light and the second-eye light into a combined light and direct the combined light to the first projection optics as the first input light.

(85) (14) The method according to (13), wherein the optical switch includes a reflective element and a first filter, and switching the optical switch to the first mode comprises: moving the reflective element to a first position, so as to reflect the first-eye light to the first filter as first-eye reflected light, and causing the first filter to reflect a first portion of the second-eye light, to transmit a second portion of the second-eye light as a first portion of the combined light, and to reflect the first-eye reflected light as a second portion of the combined light.

(86) (15) The method according to (14), further comprising: receiving a second input light by a second projector including second projection optics, wherein switching the optical switch to the second mode comprises: moving the reflective element to a second position, so as to allow the first-eye light to pass to the second projector.

(87) (16) The method according to (15), wherein the second projector includes a second filter, the method further comprising: when the optical switch is in the second mode, reflecting, by the second filter, a first portion of the first-eye light and transmitting, by the second filter, a second portion of the first-eye light as the second input light.

(88) (17) The method according to (13), wherein the optical switch includes a first filter and a reflective element, and switching the optical switch to the first mode comprises: moving the first filter to a first position, so as to allow the first-eye light and the second-eye light to pass to the first projection optics as the combined light.

(89) (18) The method according to (17), further comprising: receiving a second input light by a

second projector including second projection optics, wherein switching the optical switch to the second mode comprises: moving the first filter to a second position, so as to reflect the first-eye light and a first portion of the second-eye light to the reflective element as first-eye reflected light, and to transmit a second portion of the second-eye light as the first input light, and directing, by the reflective element, the first-eye reflected light to the second projector.

(90) (19) The method according to (18), wherein the second projector includes a second filter, the method further comprising: when the optical switch is in the second mode, reflecting, by the second filter, a first portion of the first-eye reflected light and transmitting, by the second filter, a second portion of the first-eye reflected light as the second input light.

(91) (20) A non-transitory computer-readable medium storing instructions that, when executed by a processor of a projection device, cause the projection device to perform operations comprising the method according to any one of (13) to (19).

(92) With regard to the processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. In other words, the descriptions of processes herein are provided for the purpose of illustrating certain embodiments, and should in no way be construed so as to limit the claims.

(93) Accordingly, it is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provided would be apparent upon reading the above description. The scope should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the technologies discussed herein, and that the disclosed systems and methods will be incorporated into such future embodiments. In sum, it should be understood that the application is capable of modification and variation.

(94) All terms used in the claims are intended to be given their broadest reasonable constructions and their ordinary meanings as understood by those knowledgeable in the technologies described herein unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary.

(95) The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments incorporate more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A projection system, comprising: a first light source configured to emit a first-eye light, wherein the first-eye light includes a first set of wavelengths; a second light source configured to emit a second-eye light, wherein the second-eye light includes a second set of wavelengths; a first projector including first projection optics configured to receive a first input light; and an optical

switch configured to receive all of the first-eye light and all of the second-eye light together, and switch between a first mode and a second mode, wherein the optical switch is configured to, in the first mode, combine the all of the first-eye light and a portion of the second-eye light into a combined light and direct the combined light to the first projection optics as the first input light.

2. The projection system according to claim 1, further comprising: a second projector including second projection optics configured to receive a second input light, wherein the optical switch is configured to, in the second mode, direct the all of the first-eye light to the second projection optics as the second input light.

3. The projection system according to claim 2, wherein the second projection optics includes a spatial light modulator configured to perform a spatial modulation on the second input light in response to an image data, thereby to generate a first-eye projection image when the optical switch is in the second mode.

4. The projection system according to claim 1, wherein the optical switch is configured to, in the second mode, direct the second-eye light to the first projection optics as the first input light.

5. The projection system according to claim 1, wherein the first projection optics includes a spatial light modulator configured to perform a spatial modulation on the first input light in response to an image data, thereby to generate a second-eye projection image when the optical switch is in the second mode and to generate a combined projection image when the optical switch is in the first mode.

6. The projection system according to claim 4, wherein the optical switch includes a first filter and a reflective element, and when the optical switch is in the first mode: the first filter is configured to allow the all of the first-eye light and the all of the second-eye light to pass to the first projection optics.

7. The projection system according to claim 6, further comprising: a second projector including second projection optics configured to receive a second input light, wherein, when the optical switch is in the second mode: the first filter is configured to reflect the first-eye light and a first portion of the second-eye light to the reflective element as first-eye reflected light, and to transmit a second portion of the second-eye light as the first input light, and the reflective element is configured to direct the first-eye reflected light to the second projector.

8. The projection system according to claim 7, wherein the second projector includes a second filter configured to, when the optical switch is in the second mode, reflect a first portion of the first-eye reflected light and transmit a second portion of the first-eye reflected light as the second input light.

9. The projection system according to claim 1, wherein the first light source and the second light source are laser light sources.

10. The projection system according to claim 1, wherein the first projector further includes the optical switch.

11. The projection system according to claim 1, wherein a first optical path of the first-eye light is parallel to a second optical path of the second-eye light, and wherein a second optical path of the first-eye light is not parallel to the second optical path of the second-eye light.

12. A projection system, comprising: a first light source configured to emit a first-eye light, wherein the first-eye light includes a first set of wavelengths; a second light source configured to emit a second-eye light, wherein the second-eye light includes a second set of wavelengths, wherein the second light source is distinct from the first light source; a first projector including first projection optics configured to receive a first input light; and an optical switch configured to be switched between a first mode and a second mode, wherein the optical switch is configured to, in the first mode, combine the first-eye light and the second-eye light into a combined light and direct the combined light to the first projection optics as the first input light, wherein the optical switch includes a reflective element and a first filter, and when the optical switch is in the first mode: the reflective element is configured to reflect the first-eye light to the first filter as first-eye reflected light, and the first filter is configured to reflect a first portion of the second-eye light, to transmit a

second portion of the second-eye light as a first portion of the combined light, and to reflect the first-eye reflected light as a second portion of the combined light.

13. The projection system according to claim 12, further comprising: a second projector including second projection optics configured to receive a second input light, wherein, when the optical switch is in the second mode: the reflective element is configured to allow the first-eye light to pass to the second projector.

14. The projection system according to claim 13, wherein the second projector includes a second filter configured to, when the optical switch is in the second mode, reflect a first portion of the first-eye light and transmit a second portion of the first-eye light as the second input light.

15. A method of image projection comprising: emitting a first-eye light by a first light source, wherein the first-eye light includes a first set of wavelengths; emitting a second-eye light by a second light source, wherein the second-eye light includes a second set of wavelengths; receiving, with an optical switch, all of the first-eye light and all of the second-eye light together; and switching the optical switch between a first mode and a second mode, wherein, in the first mode, the optical switch combines the all of the first-eye light and a portion of the second-eye light into a combined light and directs the combined light to a first projection optics as a first input light.

16. The method according to claim 15, further comprising: receiving a second input light by a second projector including second projection optics, wherein switching the optical switch to the second mode comprises: moving a reflective element to a second position, so as to allow the all of the first-eye light to pass to the second projector.

17. A method of image projection comprising: emitting a first-eye light by a first light source, wherein the first-eye light includes a first set of wavelengths; emitting a second-eye light by a second light source, wherein the second-eye light includes a second set of wavelengths; receiving a first input light by a first projector including first projection optics; and switching an optical switch between a first mode and a second mode, wherein the optical switch is configured to, in the first mode, combine the first-eye light and the second-eye light into a combined light and direct the combined light to the first projection optics as the first input light, wherein the second projector includes a second filter, the method further comprising: when the optical switch is in the second mode, reflecting, by the second filter, a first portion of the first-eye light and transmitting, by the second filter, a second portion of the first-eye light as the second input light.

18. The method according to claim 17, further comprising: receiving a second input light by a second projector including second projection optics, wherein switching the optical switch to the second mode comprises: moving a first filter to a second position, so as to reflect the first-eye light and a first portion of the second-eye light to a reflective element as first-eye reflected light, and to transmit a second portion of the second-eye light as the first input light, and directing, by the reflective element, the first-eye reflected light to the second projector.

19. The method according to claim 18, wherein the second projector includes a second filter, the method further comprising: when the optical switch is in the second mode, reflecting, by the second filter, a first portion of the first-eye reflected light and transmitting, by the second filter, a second portion of the first-eye reflected light as the second input light.

20. A non-transitory computer-readable medium storing instructions that, when executed by a processor of a projection device, cause the projection device to perform a set of operations comprising: controlling a first light source to emit a first-eye light, wherein the first-eye light includes a first set of wavelengths; controlling a second light source to emit a second-eye light, wherein the second-eye light includes a second set of wavelengths; controlling an optical switch to receive all of the first-eye light and all of the second-eye light together, and switch between a first mode and a second mode, wherein, in the first mode, the optical switch combines the all of the first-eye light and a portion of the second-eye light into a combined light and directs the combined light to a first projection optics as a first input light.
